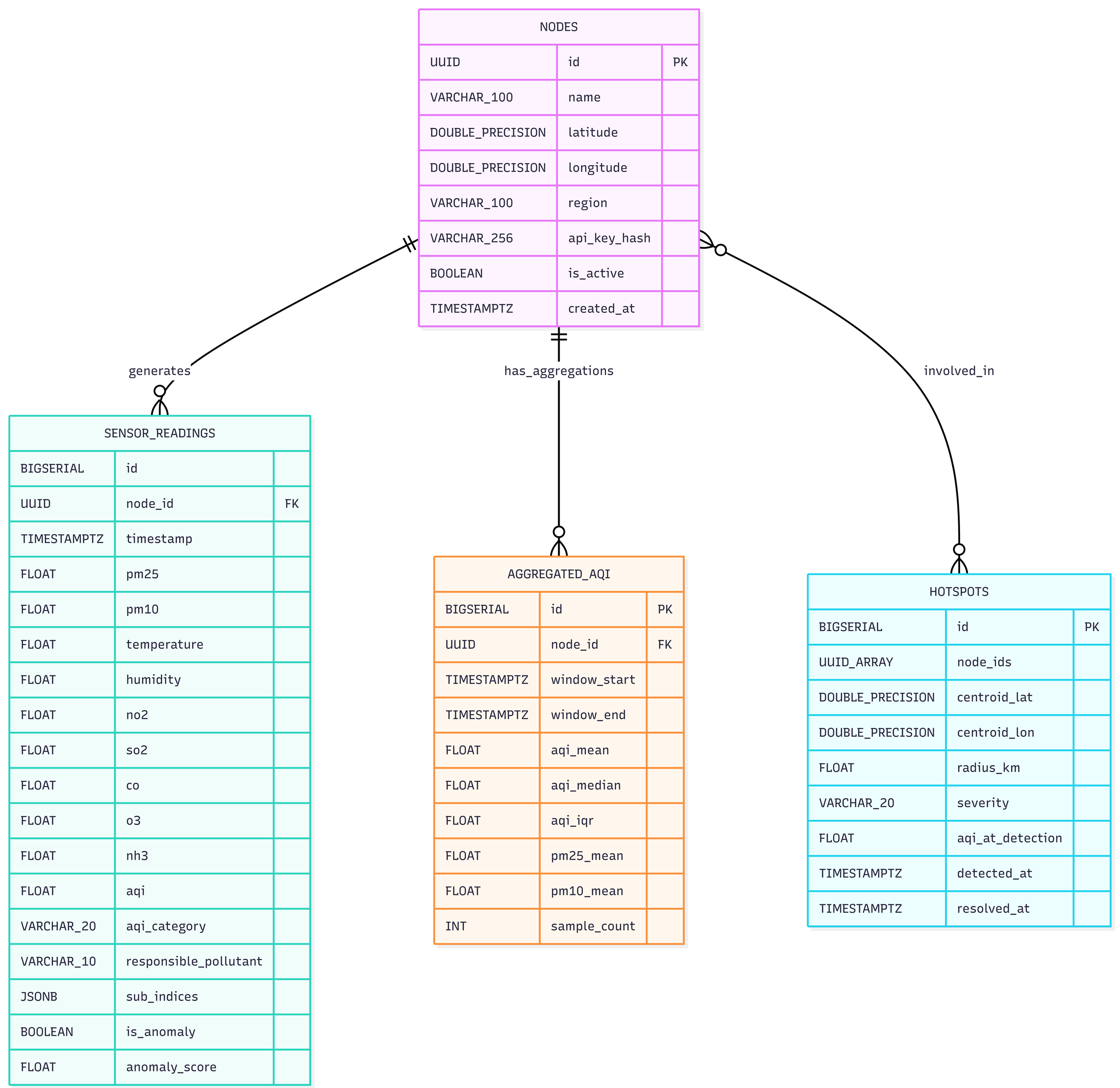

NODES
UUID id PK
VARCHAR_100 name
DOUBLE_PRECISION latitude
DOUBLE_PRECISION longitude
VARCHAR_100 region
VARCHAR_256 api_key_hash
BOOLEAN is_active
TIMESTAMPTZ created_at
generates
has_aggregations
involved_in
SENSOR_READINGS
BIGSERIAL id
UUID node_id FK
TIMESTAMPTZ timestamp
FLOAT pm25
FLOAT pm10
FLOAT temperature
FLOAT humidity
FLOAT no2
FLOAT so2
FLOAT co
FLOAT o3
FLOAT nh3
FLOAT aqi
VARCHAR_20 aqi_category
VARCHAR_10 responsible_pollutant
JSONB sub_indices
BOOLEAN is_anomaly
FLOAT anomaly_score
AGGREGATED_AQI
BIGSERIAL id PK
UUID node_id FK
TIMESTAMPTZ window_start
TIMESTAMPTZ window_end
FLOAT aqi_mean
FLOAT aqi_median
FLOAT aqi_iqr
FLOAT pm25_mean
FLOAT pm10_mean
INT sample_count
HOTSPOTS
BIGSERIAL id PK
UUID_ARRAY node_ids
DOUBLE_PRECISION centroid_lat
DOUBLE_PRECISION centroid_lon
FLOAT radius_km
VARCHAR_20 severity
FLOAT aqi_at_detection
TIMESTAMPTZ detected_at
TIMESTAMPTZ resolved_at